

	SparseAttention A	SparseAttention B	FusedAddRmsNorm A	FusedAddRmsNorm B
Expert-only: SparseAttention				
Cube Matmul for both QK and PV	—	✓	—	—
Cube / Vector stages on separate streams	—	✓	—	—
Event-driven cross-chunk pipeline	—	✓	—	—
Gather KV once; reuse across QK and PV	—	✓	—	—
Explicit L1 depth / L0 double-buffer tiling	—	✓	—	—
Expert-only: FusedAddRmsNorm				
Explicit output L2 write-cache hint	—	—	—	✓
Query hardware core count for launch grid	—	—	—	✓
Even row partition (core loads differ by <= 1)	—	—	—	✓
Shared B-side choices (not necessarily faster)				
TQue-managed producer / consumer buffers	—	✓	—	✓
ReduceSum API instead of a manual sum tree	✓*	✓	—	✓
Context: techniques also present in A				
Double buffering and input prefetch	—	✓	✓	✓
Vector arithmetic / UB reuse / mixed precision	✓*	✓	✓	✓

✓ present in selected source ✓* inherited from supplied seed — not observed / not applicable
Source-level evidence only; marks do not establish an isolated performance benefit.